Introduction to Finance
The Impact of Financing Policy on Capital Budgeting

Professor: *Víctor Macías E.*

Core project parameters & the base operating cash flow

Imagine a firm is evaluating a high-efficiency 1-year equipment upgrade project with the following financial data:

- *Initial Investment (CapEx):* \$800 at $t = 0$.
- *Expected Project EBIT:* \$1,200 at $t = 1$.
- *Corporate Tax Rate (T_C):* 25%.
- *Unlevered Cost of Capital (R_0):* 10% (reflecting pure operational/business risk).
- *Pre-Tax Cost of Debt (R_B):* 6%.

The Baseline Cash Flow

Before examining how we pay for this project, we must isolate how much cash the project generates on an all-equity, unlevered basis. Assuming no depreciation or working capital adjustments for simplicity:

$$\text{OCF} = (\text{Revenues} - \text{Expenses}) \times (1 - T_C) + \text{Depreciation} \times T_C$$
$$\text{Unlevered Free Cash Flow } (FCF_1) = \text{EBIT} \times (1 - T_C) = \$1,200 \times (1 - 0.25) = \$900$$

This \$900 represents the total operational cash "pie" available to distribute to all investors at Year 1.

1 PART 1: Constant Target Leverage Policy (Dynamic Rebalancing)

Financing Rule: The firm manages its balance sheet to maintain a strict, **constant target Debt-to-Value (B/V) ratio of 50%**.

Theoretical Foundation: Because the firm continuously rebalances its absolute dollar borrowing to stay at exactly 50% of market value, the tax shield is not safe—it fluctuates directly with business risk. If the project underperforms, value drops, and the firm must legally pay down debt (reducing the tax shield). Because the shield carries the same operational risk as the business, **the tax shield is discounted at the unlevered asset rate ($R_0 = 10\%$)**. Because the tax shield shares asset-level risk, it does not shelter equity holders from baseline business variance, causing the tax term to disappear from Modigliani-Miller Proposition II.

1.1 Method 1A: The Weighted Average Cost of Capital (WACC)

Under a target leverage policy, the overall corporate discount rate stays constant. We can back into our financing components:

1. **Find the Cost of Levered Equity (R_S):** Using MM Proposition II *without* taxes (since $B/S = 1.0$ when $B/V = 50\%$):

$$R_S = R_0 + \frac{B}{S}(R_0 - R_B) = 0.10 + 1.0 \times (0.10 - 0.06) = 14.00\%$$

2. **Calculate Corporate WACC:**

$$\text{WACC} = \frac{S}{V}R_S + \frac{B}{V}R_B(1 - T_C) = (0.50 \times 0.14) + (0.50 \times 0.06 \times [1 - 0.25]) = 9.25\%$$

3. **Value the Project (V_L):** Discounting the core operating cash flow at our blended corporate rate:

$$V_L = \frac{FCF_1}{1 + \text{WACC}} = \frac{\$900}{1 + 0.0925} = \$823.81$$

To hit the 50% target policy at t $= 0$*, the firm raises exactly* $50\% \times \$823.81 = \$411.90$ *in debt.*

1.2 Method 1B: Adjusted Present Value (APV)

1. **Unlevered Project Value (V_U):** Value if funded with 100% equity.

$$V_U = \frac{FCF_1}{1+R_0} = \frac{\$900}{1+0.10} = \$\mathbf{818.18}$$

2. **Present Value of Tax Shield ($PV(ITS)$):** Interest paid at Year 1 is $\$411.90 \times 6\% = \$24.714$. The tax shield generated is $\$24.714 \times 25\% = \$6.1785$. Because this shield tracks business risk, we discount at $R_0 = 10\%$:

$$PV(ITS) = \frac{\$6.1785}{1+0.10} = \$\mathbf{5.62}$$

3. **Total Project Value (V_L):**

$$V_L = V_U + PV(ITS) = \$818.18 + \$5.62 = \$\mathbf{823.80}$$

1.3 Method 1C: Flow-to-Equity (FTE / Capital Flow Method)

Instead of valuing the entire project, we isolate what is left over strictly for the shareholders.

1. **Calculate FCFE at $t = 1$:**

$$FCFE_1 = FCF_1 - \text{Interest} + \text{ITS} - \text{Principal Repayment}$$

$$FCFE_1 = \$900 - \$24.714 + \$6.1785 - \$411.90 = \$469.56$$

2. **Value the Equity Component (S):** Discounted at the target equity rate ($R_S = 14.0\%$):

$$S = \frac{FCFE_1}{1+R_S} = \frac{\$469.56}{1+0.14} = \$411.90$$

3. **Reconstruct Project Value:**

$$V_L = S + B = \$411.90 + \$411.90 = \$823.80$$

2 PART 2: Fixed Pre-Determined Debt Schedule

Financing Rule: The firm bypasses dynamic rebalancing and issues a contractual loan of exactly \$400 of debt at $t = 0$, regardless of operational success.

Theoretical Foundation: Because the absolute dollar schedule is permanently fixed, the corporate tax shield becomes safe, highly predictable, and entirely insulated from business shocks. Therefore, the tax shield is discounted at the safe *cost of debt ($R_B = 6\%$)*. Because this predictable tax break acts as a financial cushion, it lowers the baseline risk premium demanded by investors. Thus, the tax adjustment factor $(1 - T_C)$ is re-introduced into the MM Proposition II cost of equity formula.

2.1 Method 2A: Adjusted Present Value (APV)

Tip: When debt is explicitly fixed in dollars, start with APV to effortlessly anchor your baseline values.

1. *Unlevered Project Value (V_U):* Safely constant at $V_U = \frac{FCF_1}{1+R_0} = \frac{900}{1+0.10} = \$818.18$.
2. *Present Value of Tax Shield ($PV(ITS)$):* Fixed interest is $\$400 \times 6\% = \$24$. Fixed corporate tax shield is $\$24 \times 25\% = \$6$. Discounted at the safe contract rate $R_B = 6\%$:

$$PV(ITS) = \frac{\$6.00}{1 + 0.06} = \$5.66$$

3. *Total Project Value (V_L):*

$$V_L = V_U + PV(ITS) = \$818.18 + \$5.66 = \$823.84$$

The market value of equity is explicitly isolated as $S = V_L - B = \$823.84 - \$400 = \$423.84$.

2.2 Method 2B: Weighted Average Cost of Capital (WACC)

Given fixed initial weights $B/V = 400/823.84 = 48.55\%$ and $S/V = 423.84/823.84 = 51.45\%$, we map our variables directly:

1. *Determine the Cost of Levered Equity (R_S):* Using MM Proposition II *with* taxes to reflect the safe tax buffer:

$$R_S = R_0 + \frac{B}{S}(R_0 - R_B)(1 - T_C)$$

$$R_S = 0.10 + \left(\frac{400.00}{423.84}\right) \times (0.10 - 0.06) \times (1 - 0.25) = 12.831\%$$

2. *Calculate corporate WACC:*

$$\text{WACC} = \frac{S}{V} R_S + \frac{B}{V} R_B (1 - T_C)$$

$$\text{WACC} = (0.5145 \times 0.12831) + (0.4855 \times 0.06 \times 0.75) = 0.0660 + 0.02185 = 0.08785 \to 8.79\%$$

3. *Value the Project (V_L):*

$$V_L = \frac{FCF_1}{1 + \text{WACC}} = \frac{\$900}{1 + 0.08785} = \$827.32$$

2.3 Method 2C: Flow-to-Equity (FTE / Capital Flow Method)

1. *Calculate FCFE at $t = 1$:* Using the safe, fixed interest (\$24) and fixed tax shield (\$6):

$$FCFE_1 = (\$1200 - \$24) \times (1 - 0.25) - \$400 = \$900 - \$24 + \$6 - \$400 = \$482.00$$

2. *Value Equity (S):* Discounted at our tax-buffered equity rate ($R_S = 12.831\%$):

$$S = \frac{FCFE_1}{1 + R_S} = \frac{\$482.00}{1 + 0.12831} = \$427.19$$

3. *Total Project Value:*

$$V_L = S + B = \$427.19 + \$400.00 = \$827.19$$

2.4 Tracking the Cash Flow Pie

To master these models, look past the formulas and track where the project's physical cash flow goes at Year 1. The underlying asset creates a single \$900.00 cash pie, but capital structure determines how we slice it:

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[ CASH TO BONDHOLDERS ]                        [ CASH TO STOCKHOLDERS ]
Principal Repayment: +$411.90                  Unlevered Base FCF: +$900.00
Interest (Pre-Tax):  +$24.71                   Less Net Debt Paid: -$431.05
Less Tax Shield:     -$5.56

= NET DEBT SERVICE PAID:                       = FCFE AVAILABLE TO EQUITY:
$431.05                                        $468.95
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2.4.1 Methodology Cross-Reference Matrix

Valuation Method	Cash Flow Target	Discount Rate	Core Focus
WACC Method	Total Operational FCF_1 ($900)	WACC = 9.25% or 8.79%	**The Industry Baseline:** Safest framework when maintaining a targeted blend of debt and equity weights.
APV Method	Core Asset Value + Isolated Tax Shields	$r_U = 10\%$ (or $r_D = 6\%$ for Part 2)	**The Accounting Check:** Explicitly uncovers the precise dollar value added to the project solely by corporate tax law.
FTE Method	Residual Stockholder $FCFE_1$	r_E (Levered Equity Rate)	**The Investor View:** Tracks the pure cash return pocketed strictly by the owners of the company.

Key Lesson Takeaway: Notice that when the tax shield is completely fixed and safe (Part 2), it is discounted at a much lower rate (6% vs 10%). This generates a higher present value for your financing side effects, making total levered firm value slightly higher under a fixed debt schedule than a continuously rebalanced target structure. Always match your evaluation math to the financing rules set by management!